

KT Assignment: Airbnb Power BI Dashboard Development

Objective

Develop an end-to-end Power BI dashboard using the Airbnb dataset and incorporate advanced analytics, Python visualizations, and business insights.

Dataset

- Listings.csv
- Reviews.csv
- Data Dictionary files

Phase 1: Data Understanding & Preparation

Tasks:

1. Study the data dictionary and understand all fields.
2. Identify primary and foreign key relationships.
 - Listings ↔ Reviews using `listing_id`
3. Perform data cleaning:
 - Remove duplicates
 - Handle null values
 - Standardize date formats
 - Validate price and review metrics
4. Create a data model in Power BI.

Phase 2: Dashboard Creation

Dashboard 1: Executive Summary

KPIs :

- Total Listings
- Total Hosts
- Total Reviews
- Average Price

- Average Rating
- Super Host Count

Dashboard 2: Host Performance Analysis

Metrics :

- Host Response Rate
- Host Acceptance Rate
- Super Host Analysis
- Host Listing Count

Dashboard 3: Property & Pricing Analysis

Metrics

- Price by Property Type
- Price by City
- Room Type Comparison
- Bedroom Analysis

Dashboard 4: Review Analytics

Metrics

- Rating Distribution
- Accuracy Score
- Cleanliness Score
- Communication Score
- Location Score

Phase 3: Python Visualizations in Power BI

Create Python visuals for:

1. Price Distribution
 - Histogram
2. Correlation Matrix
 - Histogram
3. Property Type Analysis
 - Box Plot

Phase 5: Dashboard Enhancements

UI/UX Requirements

- Dynamic slicers
- Bookmarks
- Drill-through pages
- Tooltips
- Conditional formatting
- Navigation menu

Phase 6: Business Insights

Generate insights such as:

1. Which city has the highest average listing price?
2. Which room type generates better ratings?
3. What factors influence review ratings?
4. Are Super Hosts charging higher prices?
5. Which amenities impact customer satisfaction?
6. Which cities have the fastest review growth?